

Improved SAC Algorithm for Solving the Interactive Hybrid Disassembly Line Balancing Problem Considering Multiskilled Workers

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Abstract—With the advancement of manufacturing and the progression towards intelligent production, more and more production enterprises are focusing on the hybrid disassembly line balancing problem to enhance efficiency and reduce costs. This work presents an interactive hybrid disassembly line balancing problem considering multiskilled workers and its corresponding mathematical model. Additionally, we incorporated the concept of carbon emissions into the problem to ensure the green sustainability of the production process. We employ an improved soft actor-critic (SAC) algorithm for solving the problem to achieve the goals of maximizing profit, minimizing idle time, and minimizing carbon emissions. The experimental results demonstrate that the improved SAC algorithm performs well on six different experimental scale cases. Specifically, multiobjective soft actor-critic (MO-SAC) produces solutions that are closer to the Pareto front than those obtained by deep deterministic policy gradient and nondominated sorting genetic algorithm II. This

advantage is particularly pronounced in medium and large-scale problem instances. Statistical comparisons using nondominated solution metrics (spread, epsilon, and IGD+), Pareto front distributions, and computational time show that MO-SAC achieves better convergence and maintains comparable computational efficiency, highlighting its performance benefits in complex scenarios. The results verify the effectiveness of our proposed approach, highlighting the superiority of the improved SAC algorithm in addressing the multiobjective interactive hybrid disassembly line balancing problem.

Index Terms—Interactive hybrid disassembly line balancing problem (IHDLBP), multiobjective soft actor-critic (MO-SAC), multiskilled workers, sustainability.

NOMENCLATURE

$\mathbb{P}$	The set of EOL products, $\mathbb{P} = \{1, 2, \dots, p\}$.
J_p	The quantity of tasks of product p .
$\mathbb{L}$	The set of DLs.
$\mathbb{W}^1$	The set of workstations on the linear DL, $\mathbb{W}^1 = \{1, 2, \dots, W^1\}$.
$\mathbb{W}^2$	The set of workstations on the U-shaped DL, $\mathbb{W}^2 = \{1, 2, \dots, W^2\}$.
$\mathbb{S}$	The set of two sides in a U-shaped configuration, $\mathbb{S} = \{1, 2\}$.
$\mathbb{N}$	The set of disassembly skills, $\mathbb{N} = \{1, 2, \dots, N\}$.
v_{pj}	The revenue derived from performing disassembly task j for product p .
t_{pj}	The duration needed to complete disassembly task j for product p .
c_{pj}	The cost for performing disassembly task j for product p per unit time.
c_l	The cost for operating on DL l per unit time.
c_w	The cost for operating on workstation w per unit time.
C	The training cost for worker skills.
α_{lwn}	Indicates that the worker at workstation w on DL l has skill n .
β_{pjn}	Indicates that disassembly task j for product p requires skill n .
PW_w^l	Power load of workstation w on DL l .
PD_l	Power consumption of DL l .
EF_{elc}	Carbon emission coefficient.

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I. INTRODUCTION

IN recent years, the rapid growth of the electronics industry has led to a large accumulation of end-of-life (EOL) products such as electronics, automotive parts, and appliances, whose improper management poses serious environmental challenges [1], [2], [3], [4], [5]. Effective disassembly and recycling can recover valuable materials, reduce waste, and support the principles of circular economy [6], [7], [8], [9], [10], [11]. The disassembly line balancing problem (DLBP) [12], [13] focuses on assigning tasks to workstations in linear [14], [15], [16], [17] or U-shaped layouts [18], [19], [20], [21]. DLBP plays a key role in this process by minimizing idle time and costs while balancing workloads for sustainable disassembly [17], [22], [23].

We extend DLBP by proposing the interactive hybrid disassembly line balancing problem (IHDLP), combining linear and U-shaped disassembly lines (DLs) [24], [25], [26], [27] for greater flexibility and efficiency. IHDLP introduces multiple objectives—profit maximization [28], carbon emission reduction [29], and idle time minimization [30], [31], [32]—while incorporating worker skills and interactive factors. Integrated formulations combining multimanned [12], [33], [34] DL balancing, lot sizing, and reverse supply chain network design have been proposed to minimize total system costs under different distribution strategies [35]. Yeni et al. [36] focus on multiproduct and multiperiod DL planning with multimanned stations, emphasizing the coordination between long-term planning and short-term lot sizing decisions using metaheuristic approaches. Multiobjective DLBPs considering worker heterogeneity and group-based assignment strategies have been proposed at the tactical and operational levels to analyze trade-offs among cost, cycle time, and workload balance [37]. Different from conventional multimanned DLs in which workers are restricted to operating on both sides of the same line, the proposed IHDLP permits workers on one DL to execute tasks on another line, enabling dynamic cross-line interaction at the operational decision level. This hybrid structure addresses the diverse needs of modern disassembly tasks and enhances system adaptability and sustainability. To clearly position the proposed work within the existing literature, a comparative summary of representative DLBP studies is provided in Table I.

IHDLP is an NP-hard problem [38], [39], [40], where solution complexity grows rapidly with problem size. For example, Khan et al. [41] propose an enhanced beetle antennae search algorithm that integrates adaptive moment estimation to improve convergence for nonconvex problems. Liu et al. [42] propose a batch EOL products human-robot collaborative remanufacturing process planning and job-shop scheduling problem, which is formulated as a two-layer multiobjective optimization model and solved using a hybrid grey wolf optimizer and rat swarm optimizer. Yuan et al. [43] develop a simulated annealing-based bees algorithm to optimize task allocation in distributed green cloud systems, while Li et al. [44] introduce a Q-learning-enhanced artificial bee colony algorithm for scheduling problems, aiming to minimize makespan. These heuristic approaches have demonstrated effectiveness in

exploring large solution spaces. However, their performance often deteriorates as problem complexity increases, frequently yielding suboptimal results in dynamic and multifaceted scenarios such as IHDLP.

To address these challenges, we adopt reinforcement learning (RL), which allows agents to learn optimal strategies through trial and error [45], [46], [47]. RL's adaptability makes it ideal for complex combinatorial problems [48], [49], [50]. Applications span scheduling [51], salesman problems [52], finance [53], and vehicle routing [54]. We employ the SAC algorithm, which uses entropy-based exploration and stochastic policy optimization to handle continuous actions. SAC's flexibility suits the dynamic and multiobjective nature of IHDLP, aiming to improve resource recovery, cost efficiency, and sustainability.

To systematically investigate the proposed IHDLP and the applicability of RL, this work is guided by the following research questions.

- 1) How can an IHDLP considering multiskilled workers and multiple conflicting objectives be effectively formulated to reflect realistic operational characteristics?
- 2) How can an RL-based framework be designed to address the dynamic decision-making requirements of the proposed multiobjective IHDLP?
- 3) To what extent can the proposed approach improve system performance, in terms of economic efficiency, environmental impact, and operational balance, compared with existing solution methods under different problem scales?

The main contributions of this work are as follows.

- 1) *Theoretical Contributions*: We use a new DL layout to address the DLBP. In the proposed IHDLP, beyond the objective of profit maximization, we incorporate two additional objectives: minimizing carbon emissions and minimizing idle time. These objectives refine the IHDLP, making it more aligned with real-world scenarios, and enhancing both efficiency and sustainability [55], [56].
- 2) *Methodological Contributions*: We enhance the SAC algorithm and develop an RL environment tailored for IHDLP. This adaptation gives the algorithm a significant advantage in solving multiobjective problems.
- 3) *Managerial Contributions*: We formulate the mathematical model for the proposed problem and employ MO-SAC, DDPG, and NSGAI algorithms to solve it. We compare the performance of these three algorithms across different case scales using experimental results and various evaluation metrics.

II. INTERACTIVE HYBRID DISASSEMBLY LINE BALANCING PROBLEM CONSIDERING MULTISKILLED WORKERS

A. Precedence Diagram

A precedence diagram [57], [58] visually represents the dependency and execution order of tasks. It helps identify task sequences, critical paths, and scheduling constraints. Tasks are shown as nodes, with directed arrows indicating which tasks must precede others. This clear structure aids in effective planning and resource allocation. In this work, we adopt precedence

TABLE I
COMPARISON OF REPRESENTATIVE STUDIES ON DISASSEMBLY LINE BALANCING PROBLEMS

Reference	Line structure	Product	Objectives	Solution Approach
Gungor et al. [5]	Linear	Single	Number of workstations	State Network
Agrawal et al. [21]	U-shaped	Multi	Resources use, Line efficiency	Swarm Intelligence Algorithm
Özceylan et al. [15]	Linear	Single	Cost	Optimization Model
Tuncel et al. [30]	Linear	Single	Number of workstations, Idle time, Sequencing positions	Reinforcement Learning
Hezer et al. [11]	Parallel	Single	Number of workstations	Exact Algorithm
Seidi et al. [31]	Linear	Single	Number of workstations, Idle time, Sequencing positions	Genetic Algorithm
Zhang et al. [32]	Linear	Single	Cost, Number of workstations, Idle time	Swarm Intelligence Algorithm
Fang et al. [33]	Parallel(multi-manned)	Multi	Cycle time, Energy consumption, Peak workstation energy consumption	Optimization Model and Simulated Annealing
Diri et al. [34]	Linear(multi-manned)	Multi	Cost	Optimization Model
Liu et al. [12]	Linear(multi-manned)	Multi	Cost	Genetic Algorithm
Cil et al. [16]	Linear	Single	Cycle time, Number of workers	Exact Algorithm and Genetic Algorithm
Zhang et al. [17]	Linear	Single	Cost	Optimization Model and Genetic Algorithm
Yılmaz et al. [38]	Linear(multi-manned)	Single	Cost, Cycle time, Workload	Exact Algorithm
Zhang et al. [24]	Hybrid	Multi	Profit	Tabu Search Algorithm
Guo et al. [20]	U-shaped	Single	Profit, Difference in fatigue	Evolutionary Algorithm
Qin et al. [23]	Parallel	Multi	Profit	Swarm Intelligence Algorithm
Zeng et al. [69]	Linear	Single	Cycle time	Optimization Model and Simulated Annealing
Wu et al. [29]	Linear	Single	Number of workstations, Smoothing indices, Carbon emissions	Genetic Algorithm
Xu et al. [28]	Parallel	Single	Profit, Number of workstations, Number of sub workstations, Workload smoothness, Hazard index	Genetic Algorithm
Yeni et al. [37]	Linear(multi-manned)	Multi	Cost	Optimization Model and Genetic Algorithm
Öksüz et al. [36]	Linear(multi-manned)	Multi	Cost	Optimization Model
This work	Interactive hybrid	Multi	Profit, Carbon emissions, Idle time	Reinforcement learning

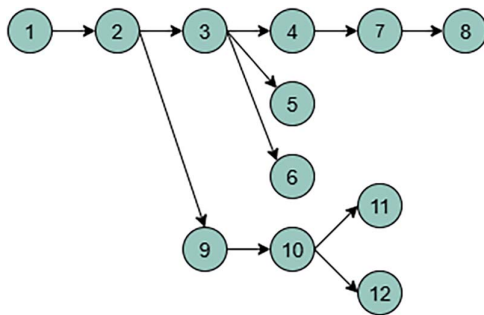


Fig. 1. Precedence diagram of a cellphone.

diagrams to illustrate the task relationships in the disassembly process of IHDLBP. Fig. 1 illustrates the task precedence relationships in the disassembly process. For instance, Task 1 must precede Task 2, which in turn enables both Task 3 and Task 9. Task 3 serves as a prerequisite for Tasks 4, 5, and 6.

Downstream, Task 4 must be completed before Task 7, which then enables Task 8. Similarly, Task 10 depends on Task 9 and leads to Tasks 11 and 12. Tasks at the same level without direct precedence relations, such as Tasks 4, 5, and 6, or Tasks 11 and 12, can be executed in parallel and have no priority order.

B. Problem Description

This work explores an IHDLBP, which integrates two types of DLs (linear and U-shaped). Both types of DLs can simultaneously disassemble multiple products, each with unique advantages. To better meet the diverse disassembly needs and improve efficiency, we integrate both DLs to address the DLBP.

During the disassembly process, different tasks may require different disassembly methods, necessitating various worker skills. For instance, disassembling complex components and handling dangerous parts requires specific expertise. Workers lacking the necessary skills must undergo additional training before performing tasks, or tasks can be assigned to workers

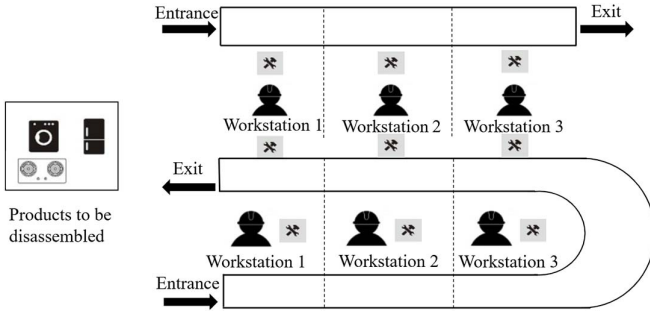


Fig. 2. Interactive hybrid disassembly line.

with the appropriate expertise. The main goal of this study is to optimize the IHDLBP by increasing profit, reducing carbon emissions, and decreasing idle time.

In a typical DLBP, workers are required to operate within their assigned workstations on their respective DLs. This means that each worker is confined to a specific location and cannot move between different lines. In the proposed IHDLBP, we consider a disassembly plant that includes a linear DL and a U-shaped DL. The workstations on the linear DL can handle tasks assigned to the linear line as well as tasks at the exit side of the U-shaped line, while the workstations on the U-shaped line can only process tasks allocated to it. An example of an interactive hybrid DL layout is shown in Fig. 2.

Note that the term interaction in the proposed IHDLBP differs fundamentally from that used in conventional hybrid, parallel, or multiconnected DL models. In traditional settings, disassembly lines with different structural configurations typically operate in a relatively independent manner, and interactions are often limited to static task allocation or predefined workload sharing. The interaction considered in this work represents a dynamic and bidirectional coupling between the linear and U-shaped DLs. Task assignment decisions on one line directly influence feasible task sets, resource availability, and precedence relations on the other line, resulting in a mutually constrained decision space. This interaction affects the structural configuration of the system and the operational decision-making process, such as worker assignment and task sequencing.

A key aspect of IHDLBP is task allocation, which involves assigning various tasks to workstations in a way that balances the line and enhances efficiency. By preallocating tasks, workstations can be equipped with multiskilled workers as needed. Additionally, long periods of inactivity on any workstation can lead to resource wastage and decreased efficiency. Thus, reasonable task scheduling is crucial to reduce idle time, which involves considering task execution times, workstation capacities, and task priorities comprehensively.

C. Mathematical Model

The following assumptions are made in this work.

- 1) Products under consideration are assumed to undergo full disassembly.

- 2) Each disassembly task must be assigned to exactly one workstation, and task precedence relationships are deterministic and known in advance.
- 3) Multiskilled workers are assumed to be available and can perform tasks within their respective skill sets without interruption.
- 4) Disassembly time and power load are assumed to be constant during the planning horizon.

1) *Notations:* Nomenclature provides the definitions of the notations used throughout the model.

2) *Decision Variables:*

$$z_{pl} = \begin{cases} 1, & \text{if product } p \text{ is assigned to DL } l \\ 0, & \text{otherwise} \end{cases}$$

$$x_{pjw}^1 = \begin{cases} 1, & \text{if workstation } w \text{ of the linear DL} \\ & \text{operates task } j \text{ for product } p \\ & (w \in \mathbb{W}^1) \\ 0, & \text{otherwise} \end{cases}$$

$$x_{pjws}^2 = \begin{cases} 1, & \text{if task } j \text{ of product } p \text{ is allocated to the } s \text{ side} \\ & \text{of U-shaped DL workstation } w \\ & (w \in \mathbb{W}^2) \\ 0, & \text{otherwise} \end{cases}$$

$$y_l = \begin{cases} 1, & \text{if DL } l \text{ works} \\ 0, & \text{otherwise} \end{cases}$$

$$u_{lw} = \begin{cases} 1, & \text{if workstation } w \text{ of DL } l \text{ works} \\ 0, & \text{otherwise} \end{cases}$$

$$\xi_{lwn} = \begin{cases} 1, & \text{if a worker on workstation } w \text{ of the disassembly} \\ & \text{line } l \text{ requires skill } n (n \in \mathbb{N}) \\ 0, & \text{otherwise} \end{cases}$$

T^l Cycling time of DL l .

3) *Task Precedence Relationship Matrix:* To represent the precedence relationships in a precedence diagram, we define a task precedence matrix $D = [d_{pjk}]$ as follows:

$$d_{pjk} = \begin{cases} 1, & \text{if task } j \text{ of product } p \text{ must be completed} \\ & \text{before task } k \\ 0, & \text{otherwise} \end{cases}.$$

4) *Mathematical Model:*

Objective function

$$\begin{aligned} \max & \sum_{p \in \mathbb{P}} \sum_{j \in \mathbb{J}_p} \left(\sum_{w \in \mathbb{W}^1} (v_{pj} - c_{pj}t_{pj}) x_{pjw}^1 \right. \\ & \left. + \sum_{w \in \mathbb{W}^2} \sum_{s \in \mathbb{S}} (v_{pj} - c_{pj}t_{pj}) x_{pjws}^2 \right) \\ & - \sum_{l \in \mathbb{L}} \left(c_l T^l + \sum_{w \in \mathbb{W}^l} c_w u_{lw} \right) \\ & - C \sum_{l \in \mathbb{L}} \sum_{w \in \mathbb{W}^l} \sum_{n \in \mathbb{N}} (\xi_{lwn} - \alpha_{lwn}) \end{aligned} \quad (1)$$

TABLE II
MODEL CONSTRAINTS

Constraint	Description
(4)	Every EOL product can only be allocated to a single DL.
(5)	If product p is allocated to DL l , then that DL must be operational.
(6)	If workstation w on DL l is operational, then the DL itself must also be operational.
(7)	Workers who utilize specific skills must have proficiency in those skills.
(8)	If task j of product p is performed at workstation w on the linear DL, then product p must also be assigned to the linear DL.
(9)	If task j of product p is executed at workstation w on the linear DL, then workstation w must be operational.
(10)	If task j of product p is carried out at workstation w on the linear DL, then the worker at workstation w must possess the necessary skills for that task.
(11)	If task j of product p is performed on the s side of workstation w on the U-shaped DL, then product p must be assigned to the U-shaped DL.
(12)	If task j of product p is executed on the s side of workstation w on the U-shaped DL, then workstation w must be operational.
(13)	If task j of product p is carried out on the s side of workstation w on the U-shaped DL, then the worker at workstation w must possess the necessary skills for that task.
(14)-(15)	The total time spent on disassembly tasks at any workstation must not exceed the cycle time of the DL.
(16)-(17)	A task is only executed after its predecessor task, and is assigned to a single workstation. If $d_{pjk} = 1$, task j must be assigned to an earlier or the same workstation as task k within the disassembly sequence of the same product.
(18)-(19)	If $d_{pjk} = 1$, task j must be assigned and completed before task k .
(20)	Each task must be performed only once.
(21)-(27)	These constraints specify the variable ranges.

$$\min EF_{elc} \sum_{l \in \mathbb{L}} \left(\left(\sum_{w \in \mathbb{W}^l} PW_w^l \cdot T^l \cdot u_{lw} \right) + PD_l \cdot T^l \right) \quad (2)$$

$$\min \left(\sum_{w \in \mathbb{W}^1} \left(\max(T^1, T^2) - \sum_{p \in \mathbb{P}} \sum_{j \in \mathbb{J}_p} x_{pjw}^1 \cdot t_{pj} \right) + \sum_{w \in \mathbb{W}^2} \left(\max(T^1, T^2) - \sum_{p \in \mathbb{P}} \sum_{j \in \mathbb{J}_p} \sum_{s \in \mathbb{S}} x_{pjws}^2 \cdot t_{pj} \right) \right). \quad (3)$$

The objective function (1) represents the maximization of the profit in IHDLBP. The first term of the equation denotes the total profit obtained from disassembling products on both linear and U-shaped DLs, the second term represents the total cost of operating the DLs and their workstations, and the third term indicates the total cost of training workers for the required disassembly skills. The objective function (2) represents the minimization of carbon emissions, calculated using the carbon emission coefficient and the total energy consumption. The objective function (3) represents the minimization of idle time.

This is calculated by subtracting the operating time of each workstation from the maximum cycle time of the DL.

Subject to

$$\sum_{l \in \mathbb{L}} z_{pl} = 1, \forall p \in \mathbb{P} \quad (4)$$

$$z_{pl} \leq y_l, \forall p \in \mathbb{P}, \forall l \in \mathbb{L} \quad (5)$$

$$u_{lw} \leq y_l, \forall w \in \mathbb{W}^l, \forall l \in \mathbb{L} \quad (6)$$

$$\xi_{lwn} \geq \alpha_{lwn}, \forall l \in \mathbb{L}, \forall w \in \mathbb{W}^l, \forall n \in \mathbb{N} \quad (7)$$

$$x_{pjw}^1 \leq z_{pl}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, l = 1, \forall w \in \mathbb{W}^1 \quad (8)$$

$$x_{pjw}^1 \leq u_{lw}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, l = 1, \forall w \in \mathbb{W}^1 \quad (9)$$

$$x_{pjw}^1 \leq \sum_{n \in \mathbb{N}} \beta_{pjn} \xi_{lwn}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, l = 1, \forall w \in \mathbb{W}^1 \quad (10)$$

$$x_{pjws}^2 \leq z_{pl}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, l = 2, \forall w \in \mathbb{W}^2, \forall s \in \mathbb{S} \quad (11)$$

$$x_{pjws}^2 \leq u_{lw}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, l = 2, \forall w \in \mathbb{W}^2, \forall s \in \mathbb{S} \quad (12)$$

$$x_{pjws}^2 \leq \sum_{n \in \mathbb{N}} \beta_{pjn} \xi_{lwn}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, l = 2, \forall w \in \mathbb{W}^2 \quad (13)$$

$$\forall s \in \mathbb{S} \quad (13)$$

$$\sum_{p \in \mathbb{P}} \sum_{j \in \mathbb{J}_p} t_{pj} x_{pjw}^1 \leq T^1, \forall w \in \mathbb{W}^1 \quad (14)$$

$$\sum_{p \in \mathbb{P}} \sum_{j \in \mathbb{J}_p} \sum_{s \in \mathbb{S}} t_{pj} x_{pjws}^2 \leq T^2, \forall w \in \mathbb{W}^2 \quad (15)$$

$$\sum_{w \in \mathbb{W}^1} w (x_{pjw}^1 - x_{pkw}^1) + W^1 \left(\sum_{w \in \mathbb{W}^1} x_{pkw}^1 - 1 \right) \leq 0 \quad (16)$$

$$\forall p \in \mathbb{P}, \forall j, k \in \mathbb{J}_p \text{ and } d_{pjk} = 1$$

$$\sum_{w \in \mathbb{W}^2} (w (x_{pjw}^2 - x_{pkw}^2) + (2W^2 - w) (x_{pjw}^2 - x_{pkw}^2)) + 2W^2 \left(\sum_{w \in \mathbb{W}^2} \sum_{s \in \mathbb{S}} x_{pkws}^2 - 1 \right) \leq 0 \quad (17)$$

$$\forall p \in \mathbb{P}, \forall j, k \in \mathbb{J}_p \text{ and } d_{pjk} = 1 \quad (17)$$

$$d_{pjk} \left(\sum_{w \in \mathbb{W}^1} x_{pkw}^1 - \sum_{w \in \mathbb{W}^1} x_{pjw}^1 \right) \leq 0 \quad (18)$$

$$d_{pjk} \left(\sum_{w \in \mathbb{W}^2} \sum_{s \in \mathbb{S}} x_{pkws}^2 - \sum_{w \in \mathbb{W}^2} \sum_{s \in \mathbb{S}} x_{pjws}^2 \right) \leq 0 \quad (19)$$

$$\sum_{w \in \mathbb{W}^1} x_{pjw}^1 + \sum_{w \in \mathbb{W}^2} \sum_{s \in \mathbb{S}} x_{pjws}^2 \leq 1, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p \quad (20)$$

$$z_{pl} \in \{0, 1\}, \forall p \in \mathbb{P}, \forall l \in \mathbb{L} \quad (21)$$

$$x_{pjw}^1 \in \{0, 1\}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, w \in \mathbb{W}^1 \quad (22)$$

$$x_{pjws}^2 \in \{0, 1\}, \forall p \in \mathbb{P}, \forall j \in \mathbb{J}_p, w \in \mathbb{W}^2, s \in \mathbb{S} \quad (23)$$

$$y_l \in \{0, 1\}, \forall l \in \mathbb{L} \quad (24)$$

$$u_{lw} \in \{0, 1\}, \forall l \in \mathbb{L}, \forall w \in \mathbb{W}^l \quad (25)$$

$$\xi_{lwn} \in \{0, 1\}, \forall l \in \mathbb{L}, \forall w \in \mathbb{W}^l, \forall n \in \mathbb{N} \quad (26)$$

$$T^l \in \mathbb{R}_+, \forall l \in \mathbb{L}. \quad (27)$$

The constraints governing the model are listed in Table II with corresponding descriptions.

Taking Fig. 2 as an example, the system consists of a linear DL and a U-shaped DL, each with three workstations. Workers

assigned to different workstations possess different skill sets. Consider a product with three disassembly tasks 1, 2, 3 and precedence relations 1-2-3. Suppose task 1 is executed at the entrance side of workstation 2 of the U-shaped DL. In this case, the assigned worker must have the required skill to perform task 1. Due to precedence constraints and the structure of the U-shaped DL, task 2 can only be executed at the same workstation or at subsequent feasible workstations, including the entrance side of workstation 3 and the exit side of workstations 3, 2, and 1.

III. IMPROVED SOFT ACTOR-CRITIC ALGORITHM FOR IHDLBP

A. Markov Decision Process

A Markov decision process (MDP) [51] is a fundamental model used in RL to describe the interaction between an agent and its environment. The agent selects actions based on the current state, receives the corresponding rewards, and transitions to new states, iteratively refining its strategy to maximize long-term cumulative rewards.

Through interactions with the environment, the agent learns to develop optimal policies to make the best decisions in each state, aiming to maximize long-term rewards.

B. Multiobjective Soft Actor-Critic Algorithm

SAC is a model-free, off-policy RL algorithm designed for continuous action spaces, combining maximum entropy RL with stochastic policy optimization. We refer to the method outlined in [59] and improve it for multiobjective problems.

SAC uses two neural networks: the actor, which learns a stochastic policy, and the critic, which evaluates state-action pair values. A key feature of SAC is entropy regularization, which balances the maximization of expected returns and entropy during policy training. Traditional RL algorithms focus on optimizing cumulative rewards, but deterministic policies may lead to repetitive actions and suboptimal performance. To address this, SAC incorporates entropy maximization along with return maximization in the objective function, encouraging diversity in action selection and improving adaptability.

The MO-SAC method designed for IHDLBP in this work is shown in Fig. 3. The agent generates actions based on the current policy, which are then encoded and decoded. By interacting with the environment, the agent collects state information, rewards, and objective values. According to the memory pool rules, valid objective values are written to the memory pool, solutions are evaluated, and corresponding rewards are output. State, reward, and action information are stored in the experience buffer, and the agent updates its policy based on this information, repeating the training process. The proposed MO-SAC framework makes SAC applicable to IHDLBP by adding a Pareto dominance-based evaluation and a state-action representation designed for the characteristics of the studied problem.

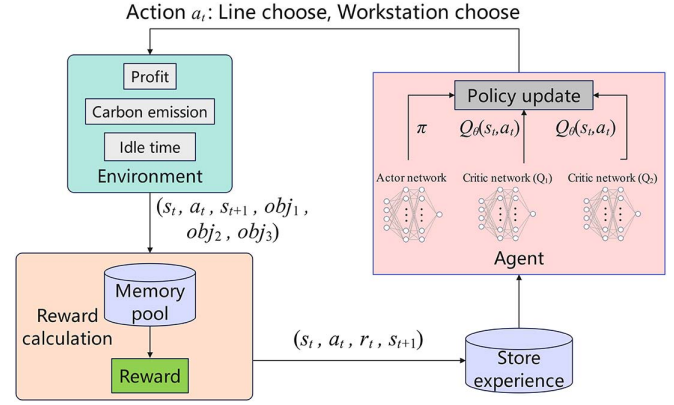


Fig. 3. MO-SAC framework for IHDLBP.

C. Environment for IHDLBP

We use OpenAI Gym to build the environment. The Box class defines a continuous action space with low and high bounds for each dimension. Compared to discrete encodings, which often lead to inefficient training or poor performance in large action spaces, the Box space allows for more precise and scalable control. Therefore, we adopt Box to represent continuous actions even in discrete problems to enhance learning efficiency.

1) *State Space Design*: The state space is primarily used to record disassembly states, which are categorized into two types: undissembled state and fully dissembled state. Initially, $S = 0$ indicates the product is in an undissembled state, while $S = 1$ signifies that the product is in a fully dissembled state after all disassembly tasks are completed. The disassembly process essentially involves transitioning the state from 0 to 1, with our focus being on the overall change in the disassembly state rather than the specific state changes in process.

This simplified state representation is adopted under the assumption of full disassembly, where decision-making focuses on system-level performance rather than intermediate process states. Workstation load is not explicitly included in the state vector, as its impact is reflected through the optimization objectives. Worker skills and task skill requirements are incorporated as feasibility constraints within the environment, where only actions that satisfy the compatibility between tasks and skills are allowed. Consequently, the state representation is designed to capture the global disassembly status.

2) *Action Space Design*: To allocate products to DLs, consider two products. We define an action space where each dimension corresponds to a product and its assigned DL, with the bounds $low = [0.0, 0.0]$ and $high = [2.0, 2.0]$. We then generate a real-valued vector of length 2, where each element ranges from 0.0 to 2.0, indicating the assignment of products to specific DLs. This range is mapped to discrete values: if a sampled value falls within $[0.0, 1.0)$, it is designated as 0, assigning the product to the linear line. If the value is within $[1.0, 2.0]$, it is designated as 1, assigning the product to the U-shaped line. For example, an action vector of $[0.3, 1.7]$ corresponds to assigning the first product to the linear DL and the second product to the U-shaped DL. The continuous actions generated by the policy network are mapped to discrete task indices

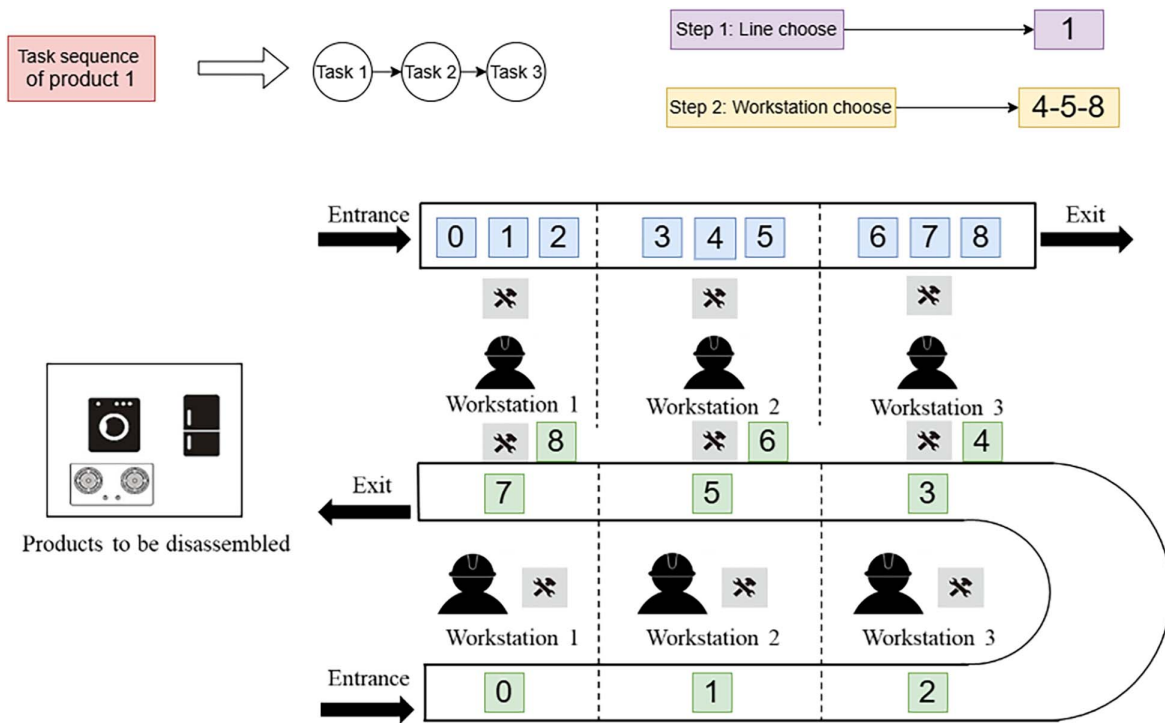


Fig. 4. Action design example. The blue squares represent the selectable actions on the linear DL (line 0). The green squares denote the selectable actions on the U-shaped DL (line 1), where actions 4, 6, and 8 indicate tasks executed at the corresponding U-shaped workstations by workers from the linear DL.

through a deterministic decoding procedure, enabling the SAC framework to handle discrete task assignment while preserving the advantages of continuous policy optimization. This design avoids defining a large discrete action space and improves learning stability when solving the combinatorial assignment problem.

For the interactive DL, we have designed a coding scheme to precisely describe the task allocation at workstations. Assuming there are three workstations on the DL, we use a numerical coding range of 0–8 to represent 12 different task allocation possibilities.

When tasks are assigned to the linear DL:

- Numbers 0–2 indicate tasks executed at workstation 1.
- Numbers 3–5 indicate tasks executed at workstation 2.
- Numbers 6–8 indicate tasks executed at workstation 3.

When tasks are assigned to the U-shaped DL:

- Number 0 represents tasks executed at the entrance side of workstation 1.
- Number 1 represents tasks executed at the entrance side of workstation 2.
- Number 2 represents tasks executed at the entrance side of workstation 3.
- Number 3 represents tasks executed at the exit side of workstation 3.
- Number 4 represents tasks executed at the underside of workstation 3 on the linear DL.
- Number 5 represents tasks executed at the exit side of workstation 2.

- Number 6 represents tasks executed at the underside of workstation 2 on the linear DL.
- Number 7 represents tasks executed at the exit side of workstation 1.
- Number 8 represents tasks executed at the underside of workstation 1 on the linear DL.

Taking Fig. 4 as an example, a product has three tasks to provide a more intuitive explanation of the disassembly process. In the figure, the choice for the type of DL is 1, indicating the product is allocated to the U-shaped DL. The workstation selections are 4-5-8, meaning task 1 is executed at the underside of workstation 3 on the linear DL, task 2 is executed at the exit side of workstation 2 on the U-shaped DL, and task 3 is executed at the underside of workstation 1 on the linear DL.

3) Reward Design: In defining the reward, we evaluate the agent's reward in this state based on the quality of the multiobjective solutions as illustrated in Fig. 5 and Algorithm 1.

First, we establish a memory pool to store multiobjective solutions during training, which is initially empty. Solutions obtained in the first training round can be successfully stored in the memory pool and receive positive feedback rewards. Whether solutions from subsequent training can enter the memory pool depends on dominance relations. The specific design of the memory pool is as follows: assuming the new solution is obj_1 , obj_2 , obj_3 , the maximum value for objective 1 in the memory pool is x , respectively; the minimum value for objective 2 is

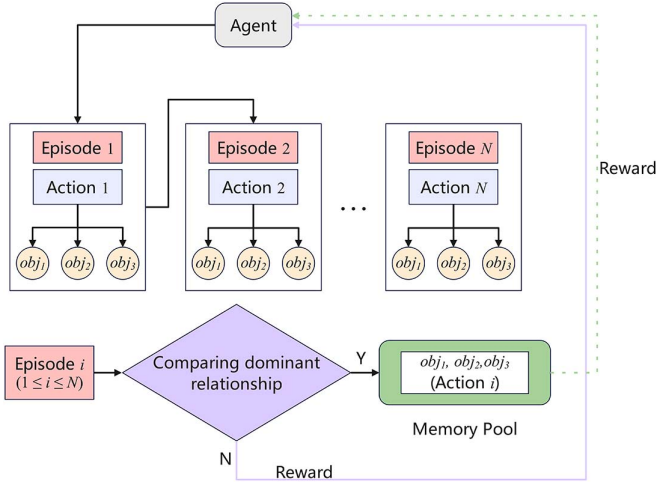


Fig. 5. Update of reward.

y , respectively; and for objective 3, the minimum value is z , respectively. The algorithm initializes the memory pool D . During the learning process, the reward is designed to encourage exploration of solutions that improve at least one of the three objectives. Specifically, if a new solution improves any of the objectives compared to the current best values in the memory pool D (obj_1 is greater than x , or obj_2 is less than y , or obj_3 is less than z), the reward is set to 100. Conversely, if the solution does not improve any of the objectives, the reward is set to -100 . This reward design ensures that the agent is incentivized to focus on finding Pareto-optimal solutions while avoiding suboptimal or redundant solutions. The memory pool D is updated whenever a new dominant or nondominant solution is discovered, which is described in the figure as the comparing dominant relationship process. This memory pool design partially handles conflicts among objectives by encouraging solutions that improve at least one objective without requiring deterioration in others. It provides a lightweight mechanism to guide the agent toward balanced improvements across multiple objectives.

The memory pool employed in this work serves a function similar to the external archive commonly used in classical multiobjective optimization methods. However, unlike traditional archive mechanisms that operate independently of the optimization dynamics, the proposed memory pool is tightly integrated with the RL process. Specifically, Pareto-optimal stored solutions directly influence reward evaluation and policy updates, thereby guiding exploration and learning in a data-driven manner. The proposed mechanism can be viewed as a learning-integrated extension of classical archive concepts.

IV. EXPERIMENTS

We apply different algorithms to the same disassembly cases and conducted a comparative analysis of the results. To verify the superiority of the MO-SAC algorithm in solving the three objectives, we compare it with the DDPG [60] and NSGAII [61] algorithms. DDPG is an RL algorithm designed for

Algorithm 1: Reward Calculation.

Input: New solution $[obj_1, obj_2, obj_3]$, memory pool D

Output: Reward r

- 1: Extract current best values from D : $x = \max(obj_1)$, $y = \min(obj_2)$, $z = \min(obj_3)$
- 2: **if** $obj_1 \geq x$ **or** $obj_2 \leq y$ **or** $obj_3 \leq z$ **then**
- 3: $r \leftarrow 100$
- 4: Add solution to memory pool D
- 5: **else**
- 6: $r \leftarrow -100$
- 7: **end if**
- 8: **return** r

TABLE III
KEY REINFORCEMENT LEARNING
PARAMETERS

Parameter	SAC	DDPG
Learning rate	0.0003	0.001
Buffer size	1 000 000	1 000 000
Batch size	256	256
τ	0.005	0.005
γ	0.99	0.99

TABLE IV
INFORMATION OF PRODUCTS

Product	Num of Tasks	Number of Required Skills
Television	12	5
Cellphone	12	5
Tesla Battery	37	5

TABLE V
CASES FOR EXPERIMENTS

Case ID	Product			Num of Tasks
	Television	Cellphone	Tesla Battery	
1	1	1	0	24
2	1	2	0	36
3	0	1	1	49
4	1	1	1	61
5	1	2	2	110
6	2	2	3	159

continuous action spaces, combining actor-critic structure with deterministic policy gradients and target networks to achieve stable convergence in high-dimensional problems. NSGAII is a classical multiobjective evolutionary algorithm that employs fast nondominated sorting and crowding distance mechanisms to approximate the Pareto front effectively, and has been widely applied in engineering optimization problems due to its robustness and efficiency. RL algorithms are implemented and

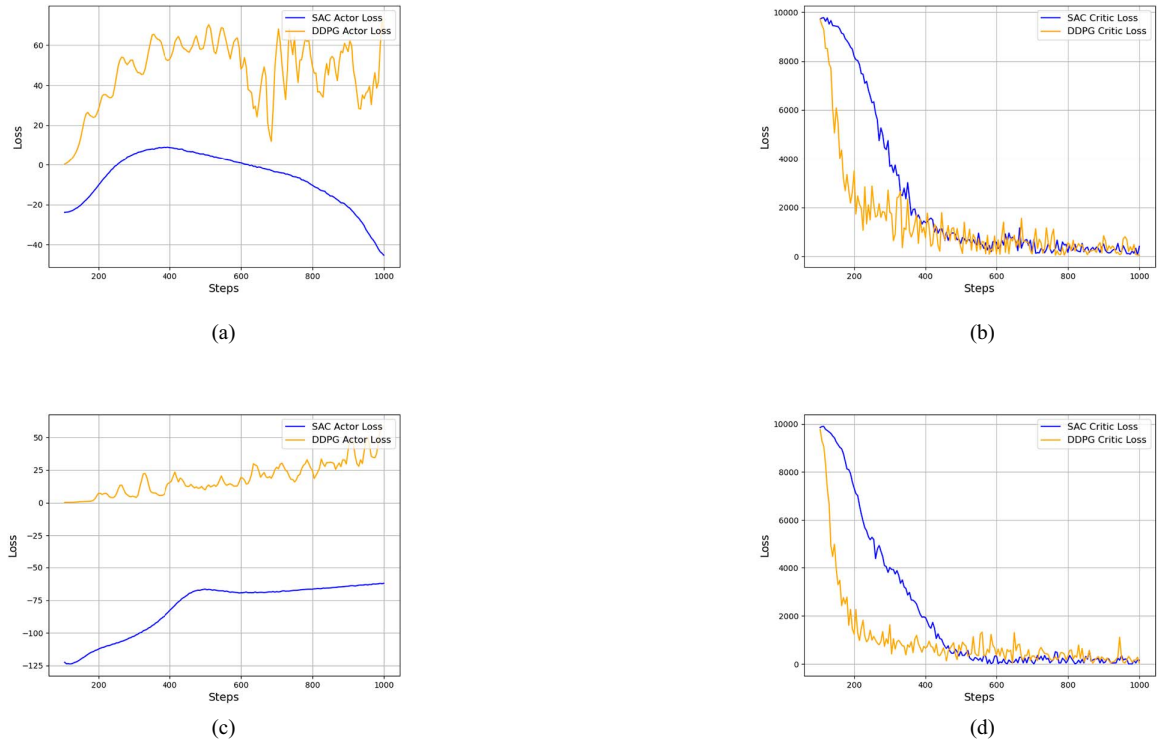


Fig. 6. Loss in training process. (a) Actor loss of Case 2. (b) Critic loss of Case 2. (c) Actor loss of Case 5. (d) Critic loss of Case 5.

TABLE VI
METRICS OF THREE ALGORITHMS FOR SOLVING MULTI-OBJECTIVE IHDLPB

Case ID	Algorithms	Spread		Epsilon		IGD+	
		Mean	t-Test	Mean	t-Test	Mean	t-Test
1	MO-SAC	0.705670321	/	0.287671233	/	0.162289957	/
	DDPG	0.550417297	+	0.17752443	-	0.101611066	-
	NSGAII	0.750555864	-	0.073289902	-	0.023969192	-
2	MO-SAC	0.702722429	/	0.259894459	/	0.032092517	/
	DDPG	0.84952051	-	0.154639175	-	0.092739948	+
	NSGAII	0.853464644	-	0.819261214	+	0.392509246	+
3	MO-SAC	0.545147404	/	0.034482759	/	0.005747126	/
	DDPG	0.969808914	-	0.372093023	+	0.326922089	+
	NSGAII	0.738668241	-	0.331597222	+	0.208876438	+
4	MO-SAC	0.495973448	/	0.060402685	/	0.008628955	/
	DDPG	0.772366368	-	0.658008658	+	0.425560618	+
	NSGAII	0.862098457	-	2.341991342	+	1.808148028	+
5	MO-SAC	0.623290576	/	0.25	/	0.067736509	/
	DDPG	0.671499919	-	1.392857143	+	1.309292611	+
	NSGAII	0.862098457	-	2.341991342	+	1.808148028	+
6	MO-SAC	0.760635265	/	0.605263158	/	0.101511341	/
	DDPG	0.648797597	+	1.078947368	+	0.721194706	+
	NSGAII	0.672350548	+	0.843298969	+	0.594289455	+

trained using the stable-baselines3 framework [62] with default parameter settings provided by the framework. For the NSGAI algorithm, the initial population is set to 50, with 200 iterations, using single-point mutation and default SBX crossover methods. We summarize the key RL parameters from the [62] in Table III. The experiments are conducted in a Windows 10 environment with an Intel(R) Core(TM) i7-7300HQ CPU (2.50 GHz) and 16.00 GB RAM.

A. Experimental Cases

We select television, cellphone [63], and Tesla battery [64] as the disassembly products in our experimental cases. Television and cellphone represent small-scale disassembly products, while the Tesla battery represents a large-scale disassembly product. We assume the maximum number of skills required to perform product disassembly tasks is 5, with details of each product shown in Table IV. We use an interval random distribution to determine the workers' disassembly skills and task profit parameters. Each disassembly case comprises different types of products. The specifics of each case are detailed in the Table V.

The number of disassembly tasks varies across products and directly influences the computational complexity. A larger number of tasks leads to a more complex state and action space, thereby increasing the difficulty of solution exploration.

B. Training Process

For each case, due to the characteristics of the reward function, the training process exhibits fluctuations in rewards and eventually stabilizes as the solution space becomes saturated, and no further dominant or nondominant solutions are identified. We present the loss functions in the training process for Cases 2 and 5 as examples in Fig. 6 (the blue line represents the SAC, the orange line represents the DDPG). In Case 2, the actor loss of SAC initially increases due to extensive exploration but gradually decreases as the policy network converges to a better strategy. However, slight fluctuations remain in the later stages due to the stochastic exploration characteristic of SAC. In contrast, the actor loss of DDPG shows significant fluctuations during training, lacking a clear convergence trend. For the critic loss, SAC exhibits a smooth and steady decline from a high initial value, reflecting a stable optimization process, whereas DDPG decreases rapidly but with greater oscillations, indicating instability. In Case 5, the actor loss of SAC initially increases and eventually stabilizes, demonstrating a smoother and more controlled optimization process. Meanwhile, the actor loss of DDPG remains highly volatile throughout the training process, failing to converge effectively. The critic loss trends are consistent with those observed in Case 2. Therefore, SAC demonstrates greater robustness and stability in complex environments. DDPG tends to exhibit instability during training and struggles to achieve effective convergence due to its deterministic policy.

C. Evaluation Metrics

For evaluating the experimental results in this work, we use the following metrics.

- 1) *Spread*: This metric measures the distribution range of the Pareto front. The value ranges from 0 to 1, where a value closer to 1 indicates better diversity and uniformity of the solution set, and a value near 0 suggests poorer diversity and uniformity.
- 2) *Epsilon*: This metric evaluates the distance between the approximate Pareto front and the true Pareto front. A smaller epsilon value suggests a closer alignment of the approximate Pareto front with the true Pareto front, indicating enhanced coverage of the solution set by the algorithm. An Epsilon value of 0 denotes complete overlap between the approximate Pareto front and the true Pareto front, indicating the discovery of the optimal solution.
- 3) *IGD+*: This metric measures the dissimilarity between the approximate Pareto front and the true Pareto front, offering insights into the quality of the solution set obtained. A diminished IGD+ value indicates superior approximation quality.

To minimize the possibility of random experimental results, we employ the t-test method in this study. In the tables, a “+” indicates that MO-SAC outperforms the comparison algorithm, while a “-” indicates that MO-SAC is less effective than the comparison algorithm. Table VI presents the specific metrics of nondominated solutions for solving the multiobjective HDLBP using three different algorithms.

D. Experimental Results and Analysis

Fig. 7 illustrates the Pareto front across different cases.

In small-scale cases, NSGAI performs the best, being closest to the true Pareto front. MO-SAC ranks second, with some solutions on the Pareto front, while DDPG performs poorly. In medium and large-scale cases, MO-SAC outperforms the others, with most solutions lying on the Pareto front. This is evident both from the data and the visuals, showing that MO-SAC solutions are closest to the true Pareto front. NSGAI performs second best, with a few solutions on the Pareto front, while DDPG continues to perform poorly.

For small-scale problems, without a specific design of crossover and mutation for NSGAI, it effectively handles low-dimensional and simple search spaces due to its search strategy, resulting in solutions close to the true Pareto front. However, in larger-scale problems, as the search space becomes more complex and extensive, MO-SAC better explores and utilizes this complexity. In the IHDLBP with three objectives, the use of MO-SAC demonstrates more pronounced advantages on medium and large scales.

Fig. 8 shows box plots of different metrics for multiple samples in Case 6. The box plots indicate that while the DDPG algorithm offers better solution diversity, the solutions generated by MO-SAC are closest to the Pareto front.

E. Computational Time Analysis

As shown in Table VII, MO-SAC and DDPG require significantly less running time than NSGAI, remaining below 21 s in all cases, while NSGAI consistently exceeds 300 s. With

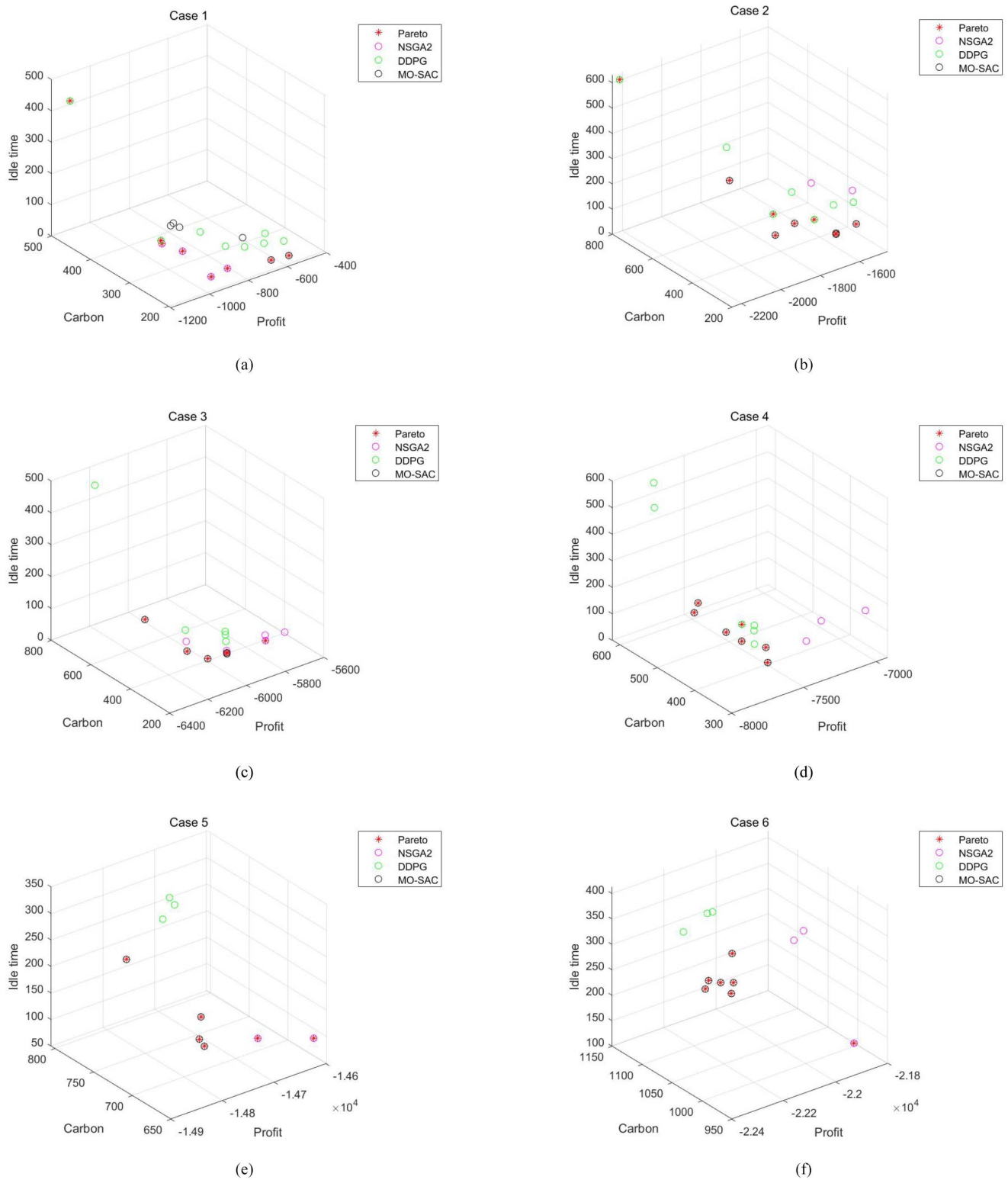


Fig. 7. Pareto front of the different cases. (a) Case 1. (b) Case 2. (c) Case 3. (d) Case 4. (e) Case 5. (f) Case 6.

increasing case complexity, MO-SAC and DDPG exhibit moderate growth in computation time, whereas NSGAII remains relatively stable due to its fixed evolutionary process, but at a

much higher cost. The results show the superior efficiency and scalability of RL methods compared to the traditional evolutionary approach.

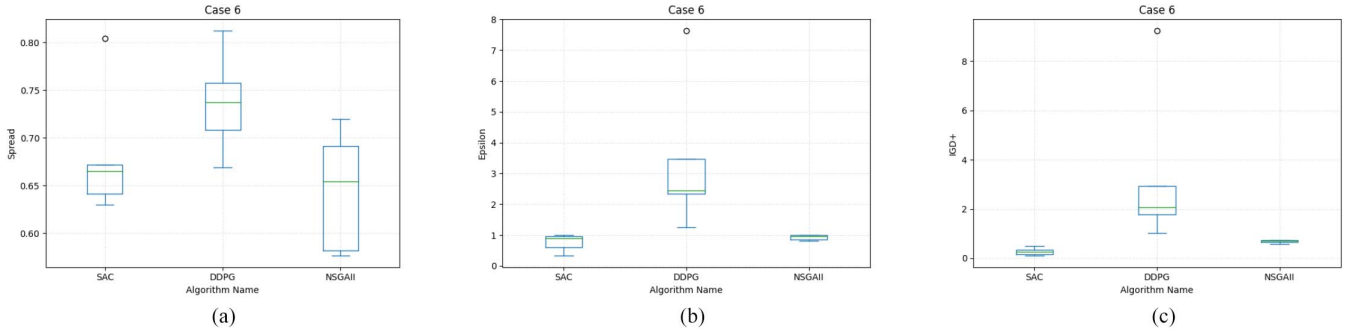


Fig. 8. Metrics of Case 6. (a) Spread indicators. (b) Epsilon indicators. (c) IGD+ indicators.

V. FINDINGS AND DISCUSSION

A. Findings

The experimental results reveal several consistent patterns. The proposed MO-SAC framework maintains stable optimization behavior across different problem scales and objective combinations, indicating significant robustness and scalability. The multiobjective outcomes exhibit clear trade-off relationships among economic profit, carbon emissions, and idle time, confirming the necessity of a Pareto-based decision-making framework rather than single-objective optimization. In the proposed learning framework, the binary reward mainly provides directional guidance for policy improvement, while the trade-off among multiple objectives is primarily reflected through Pareto dominance evaluation during the learning process.

In addition, the interaction mechanism between linear and U-shaped DLs plays a crucial role in shaping the solution space. By dynamically coupling task assignment and resource allocation decisions, the proposed model enables more balanced system-level performance. From a computational perspective, the observed runtime growth remains at an acceptable level for practical disassembly planning, supporting the applicability of the proposed approach to realistic scenarios.

B. Discussion

The experimental results across different problem scales indicate that the proposed MO-SAC framework exhibits stable performance trends as system complexity increases. Variations in objective trade-offs among economic profit, carbon emissions, and idle time lead to smooth and interpretable changes in solution distributions rather than abrupt performance degradation, a further balanced multiobjective learning behavior could be explored. Moreover, the performance implies that the effectiveness of the proposed approach is primarily driven by the problem structure and the Pareto dominance mechanism. These demonstrate a certain degree of robustness of the proposed framework under varying problem scales.

To ensure reproducibility and avoid subjective bias introduced by extensive parameter tuning, both the RL algorithm and NSGAII are implemented using commonly adopted default parameter settings reported in the literature. No problem-specific operators or parameter adjustments are applied to either

APPENDIX

TABLE VII
COMPARISON OF AVERAGE RUNNING TIME (S)
FOR DIFFERENT ALGORITHMS

Case ID	MO-SAC	DDPG	NSGAII
1	12.78	13.09	315.52
2	17.48	13.98	318.03
3	18.12	15.12	320.37
4	18.89	15.43	322.76
5	20.65	17.27	325.14
6	20.74	18.60	327.52

method. While such configurations may not fully exploit the maximum performance potential of each algorithm, they provide a neutral and transparent basis for comparison.

VI. CONCLUSION

To solve the IHDLBP, we propose a multiobjective model and evaluate MO-SAC, NSGAII, and DDPG across varying problem scales. Interactive DLs enhance flexibility and efficiency by enabling dynamic task allocation. Experimental results show that while NSGAII performs well on small instances, MO-SAC consistently outperforms other methods in larger, more complex scenarios. Its ability to adapt through RL makes it well-suited for IHDLBP, validating its effectiveness in optimizing sustainability, efficiency, and profitability. The proposed framework can be extended to other industrial disassembly operations involving dynamic task allocation, such as systems with varying product structures, flexible worker assignment, or heterogeneous workstation configurations.

From a managerial perspective, this work offers valuable insights for decision-makers in designing green disassembly systems. The results highlight the value of interactive DL structure and adaptive decision-making mechanisms in balancing economic cost, environmental impact, and operational efficiency, particularly for complex disassembly scenarios.

Despite the positive results, this work has several limitations. The proposed model relies on a set of simplifying assumptions regarding task precedence, processing times, and

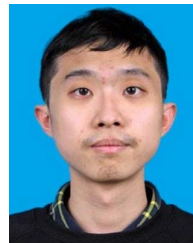
worker availability, which may not fully capture all uncertainties in real-world disassembly systems. The state representation and reward design are intentionally simplified to ensure learning stability and computational tractability, which may limit the expressive power of the RL framework in more complex environments.

In our future work, we plan to explore more fine-grained reward strategies that grade the degree of improvement for each objective. While the current binary reward mechanism is adopted to ensure stable and efficient learning in a multiobjective setting, more refined reward formulations could provide more informative feedback to the agent and potentially enhance learning efficiency and solution quality. More advanced algorithms [65], [66], [67] will be used to solve the studied problem.

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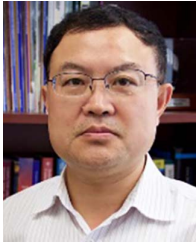
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